Penalization and Regularization

Math 742

2/5/2026

Introduction

For years ordinary least squares (OLS) regression has been the "gold standard" for regression problems. Ever since the Gauss-Markov theorem which states that the coefficients from OLS are B-L-U-E.

B - Best, of all other methods the OLS coefficients have the smallest variance of all linear unbiased coefficients. This does NOT mean OLS has low variance in absolute terms—only relative to other unbiased linear estimators.

L - Linear, the coefficients are linear. No β^2, e^β.

U - Unbiased

E - Estimates

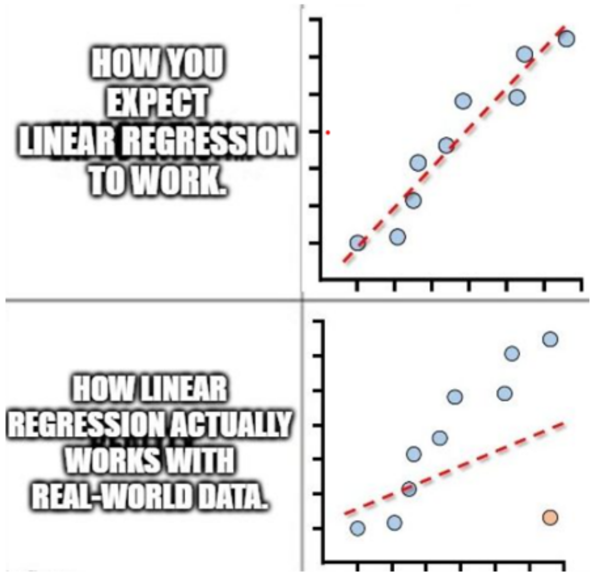
OLS regression: $\mathbf{y} = \mathbf{X}\beta + \varepsilon$ works well when,

- $n >> p$, i.e. number of observations (n) is much larger than the number of predictors (p)
- the predictors are weakly correlated

OLS doesn't work so well when,

- You have strong multicollinearity.
- p is large relative to n
- high variance, which leads to overfitting

Also, with regards to the bias variance tradeoff, OLS typically has low bias but can have high variance in these settings. Regularization (this topic) will add to the bias in order to reduce the variance.



Penalized Optimization

The basic idea is that we modify the loss function. For OLS the objective is:

$$\min_{\beta} ||\mathbf{y} - \mathbf{X}\beta||_2^2$$

$$\mathbf{X} \in \mathbb{R}^{n \times p}, \beta \in \mathbb{R}^p$$

n is the number of samples, p is the number of predictors

For penalized regression, we add a penalty term:

$$\min_{\beta} ||\mathbf{y} - \mathbf{X}\beta||_2^2 + \lambda \cdot \text{Penalty}(\beta)$$

For OLS, $\lambda = 0$. The higher the penalty, the more bias, but smaller variance. Overall, one can view penalized regression either as adding a penalty to the loss or as restricting the allowable coefficient vectors.

Ridge Regression (L2 Penalty)

Penalty

$$\sum_{j=1}^{p} \beta_j^2 = ||\beta||_2^2$$

This makes the loss function:

$$\min_{\boldsymbol{\beta}} \left\{ \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 + \lambda \sum_{j=1}^{p} \beta_j^2 \right\}$$

Key properties:

- Shrinks coefficients towards 0
- Never sets coefficients exactly to zero
- Handles multimulticollinearity
- Closed form solution exists

$$\hat{\beta}_{\text{ridge}} = (X^\top X + \lambda I)^{-1} X^\top \mathbf{y}$$

λI stabilizes the inversion with $X^T X$ is singular (or nearly singular).

→ The constraint region is circular. Why?

$$\sum_{j=1}^{p} \beta_j^2 < t \iff ||\beta||_2 < \sqrt{t}$$

For 2 predictors, $\beta_1^2 + \beta_2^2 < t$

→ The predictors must be standardized before you apply ridge. Why?

Because you want each coefficient to have "equal" effect on y. The penalty depends on the magnitude of β_j, which is scale-dependent. Without standardization, variables measured on larger scales are penalized less.

Suppose one predictor is measured in feet and another in inches. The same one unit change would have differing effects on y just because the units are different. Standardizing eliminates this.

- The penalty in the loss function depends only on β_j
- The penalty doesn't account for changes in scale in X_j, thus we need to standardize.

Lasso Regression (L1 Penalty)

Penalty

$$\sum_{j=1}^{p} |\beta_j| = ||\beta||_1$$

This makes the loss function:

$$\min_{\boldsymbol{\beta}} \left\{ \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 + \lambda \sum_{j=1}^{p} |\beta_j| \right\}$$

Key Properties

- Shrinks coefficients
- Sets some coefficients exactly to zero
- Performs variable selection
- No closed-form solution
- Solved via coordinate descent (an iterative optimization algorithm that minimizes a multivariate objective function by optimizing one parameter (coordinate) at a time, keeping all others fixed)

→ Diamond-shaped constraint

Ridge keeps all predictors on a diet. Lasso kicks some of them out of the house.

Ridge vs Lasso: Side-by-Side

Feature	Ridge	Lasso
Penalty	L_2	L_1
Shrinkage	Yes	Yes
Exact zeros	No	Yes
Correlated predictors?	Excellent	Lasso may arbitrarily select one predictor from a group of highly correlated variables.[1]
Variable selection	No	Yes
Closed form	Yes	No

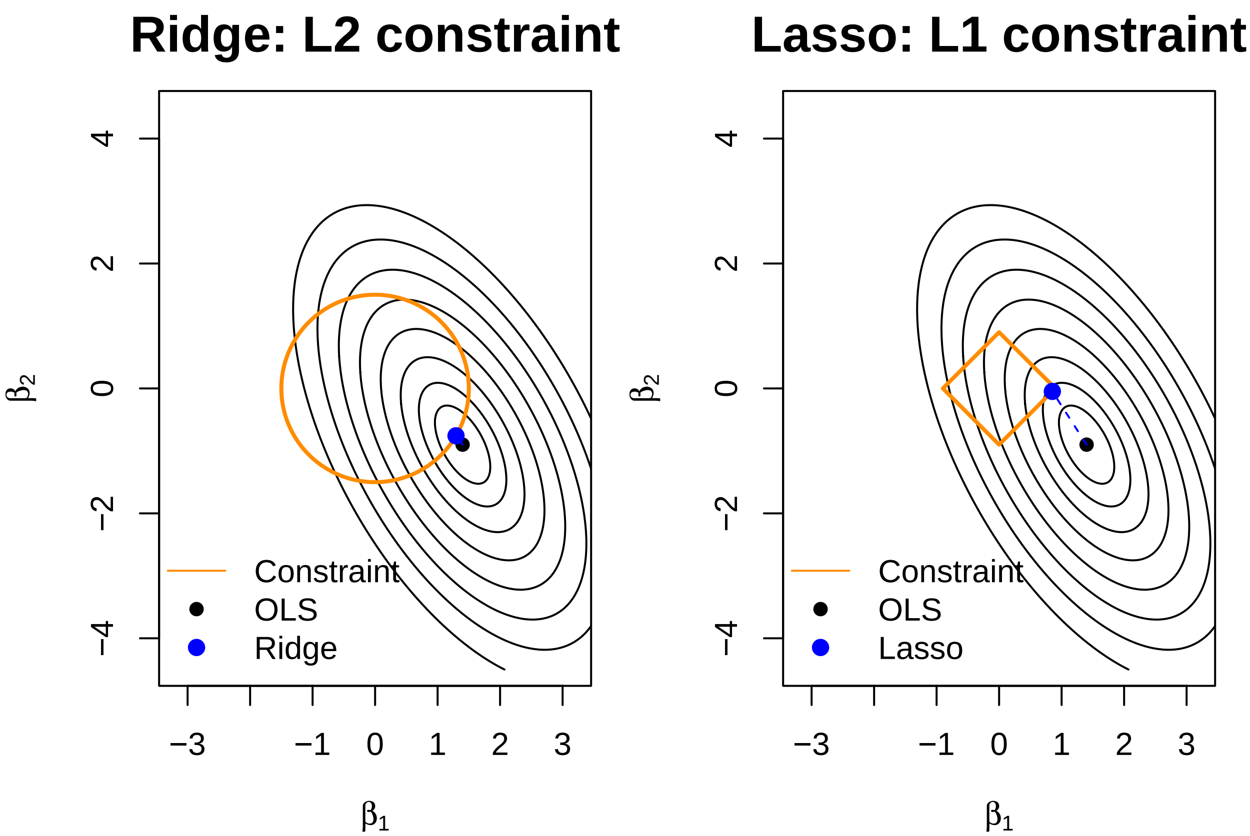
- Ridge is preferred when prediction accuracy is the goal (especially with correlated predictors).
- Lasso is preferred when model interpretability and variable selection are the goal.

Visual Comparison of Ridge vs. Lasso

To understand regularization, we can view regression geometrically. Each point (β_1, β_2) represents a possible set of coefficients, and the ellipses show values of the residual sum of squares (RSS).

Explanation of the plot:

- Each point = candidate coefficient vector (β_1, β_2).
- Ellipses = same RSS. Smaller ellipse $\rightarrow$ better fit.
- OLS = center of the smallest ellipse (minimal value of RSS).
- Regularization = restrict to constraint region
- Solution = first point where ellipse touches constraint
- Ridge (circle) $\rightarrow$ shrinkage, no zeros
- Lasso (diamond) $\rightarrow$ corners $\implies$ some $\beta_j = 0$




```
## OLS solution:    1.4 -0.9
## Ridge solution:  1.294 -0.758
## Lasso solution:  0.852 -0.048
```

In order to determine the size of the penalty λ we use cross-validation.

1. Standardize the predictors.
2. Fit the model over a grid of λ values.
3. Use K-fold cross-validation – we minimize estimated test error (prediction error), not training error.
4. Select λ that minimizes the CV error.

$$\boxed{\begin{array}{c}\lambda = 0 \rightarrow \text{OLS} \\ \lambda \rightarrow \infty \Rightarrow \text{coefficients} \rightarrow 0\end{array}}$$

Lambda selection visulization

Left Plot (CV error vs. log(λ))

- Minimum point = best predictive performance (λ_{min})
- Dotted line = λ_{1se}
- λ_{1se} is the largest λ within one standard error of the minimum CV error (simpler model with similar performance)

Right Plot: coefficient paths

- Each curve = one regression coefficient.
- $\lambda \approx 0 \rightarrow$ coefficients $\approx$ OLS
- As λ increases $\rightarrow$ coefficients shirnk towards 0

Lasso Behavior

- Some coefficients become exactly 0 $\rightarrow$ variable selection

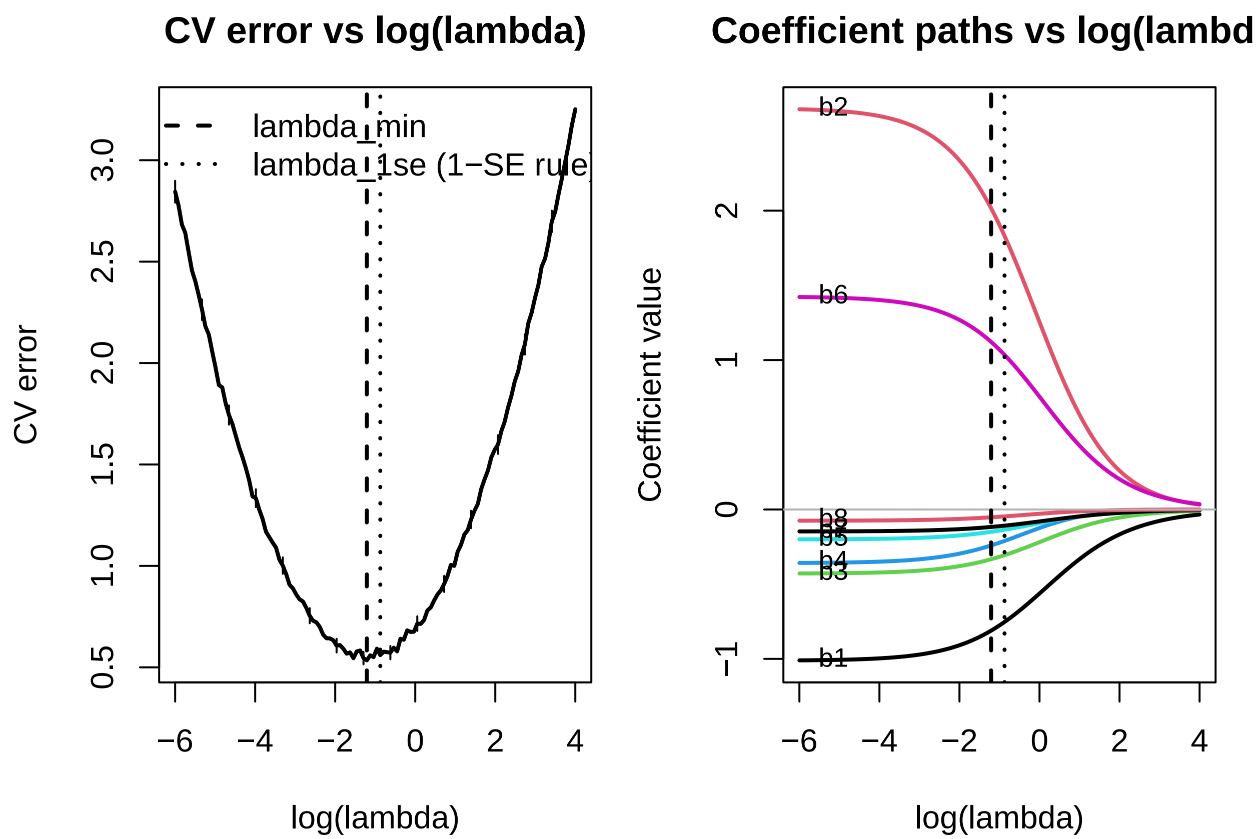
Ridge Comparison

- Same idea, but coefficients shrink smoothly
- Never exactly 0

Why plot (log(λ))?

- Spreads values out for visualization
- Example: $-6, -4, ...$ correspond to $10^{-6}, 10^{-4}$

Note: λ_{min} gives us the best fit, but λ_{1se} is almost as good, but simpler and more stable.



Bayesian Interpretation

You can frame things from a Bayesian perspective.

- Ridge uses a Gaussian prior:

$$\beta_j \sim N(0, \tau^2)$$

- Lasso uses a Laplace prior:

$$\beta_j \sim \text{Laplace}(0, b)$$

Then,

- Mathematically, the posterior mode (MAP) is the Ridge estimator if a normal prior is used and the MAP is a Lasso estimator if the Laplace prior is used.
- L2 penalty $\iff$ Gaussian prior $\implies$ smooth shrinkage.
- L1 penalty $\iff$ Laplace prior $\implies$ sharp peak at 0 $\implies$ sparsity.

Simulation

Setup

- Simulate data with highly correlated predictors

$$x1 = \varepsilon \qquad ; \qquad x_2 = x1 + \varepsilon$$

$$\varepsilon \sim N(0, \sigma = 0.05)$$

- True model uses both variables

$$y = 3x_1 + 3x_2 + \varepsilon$$

Compare:

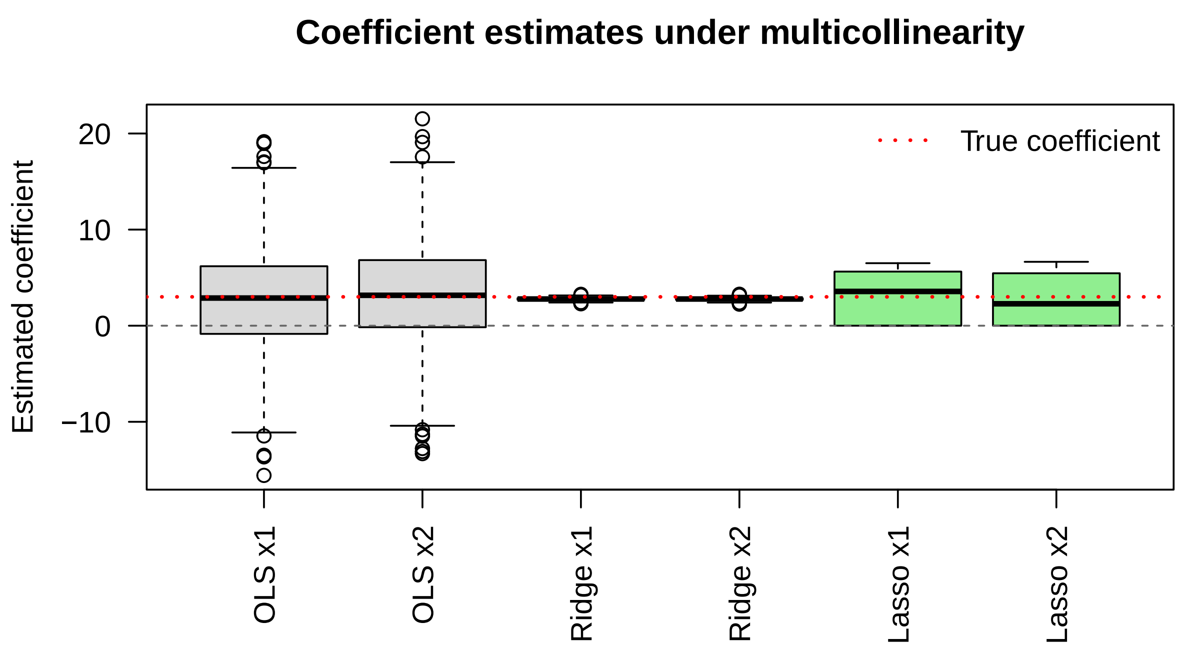
- OLS
- Ridge
- Lasso

What you should see:

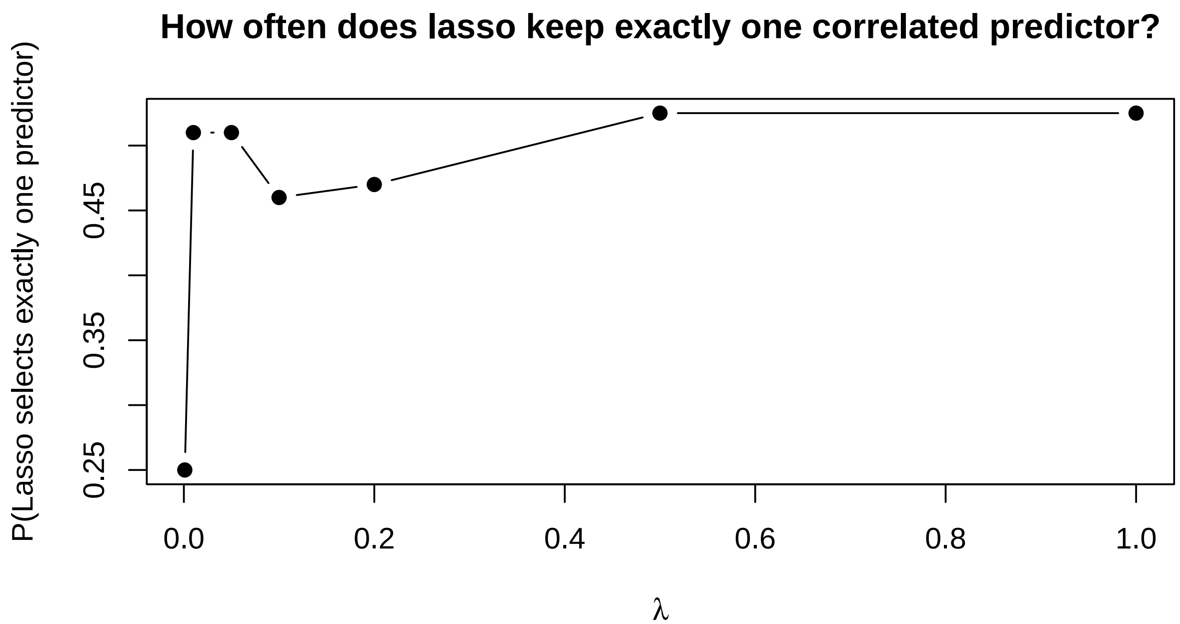
- OLS: high variance, unstable coefficients
- Ridge: shrinks together, stable
- Lasso: picks one, unstable selection (which one lasso keeps can vary across samples)

Comparison of Methods (1000 simulation runs)

```
##   Method Mean_x1 Mean_x2 SD_x1 SD_x2
## 1    OLS   2.785   3.219 5.284 5.284
## 2  Ridge   2.781   2.773 0.140 0.141
## 3  Lasso   3.131   2.669 2.450 2.448
```



Lasso Coefficient Summary



```
##              lambda prob_exactly_one_selected
## lambda_0.001  0.001                     0.250
## lambda_0.01   0.010                     0.510
## lambda_0.05   0.050                     0.510
## lambda_0.1    0.100                     0.460
## lambda_0.2    0.200                     0.470
## lambda_0.5    0.500                     0.525
## lambda_1      1.000                     0.525
```

- Notice how OLS coefficients are all over the place. That's multicollinearity → high variance
- Ridge keeps both variables and shrinks them together. Much more stable.
- Lasso often picks one and kills the other—even though both matter. And which one it picks can change.
- This is not because one predictor is better, it's because they contain nearly identical information.

$\boxed{\text{With correlated predictors: OLS is unstable, Ridge is stable, Lasso is selective but unstable.}}$

2 Questions

1. Which model would you use if prediction accuracy was the main goal?

2. Which model would you use if interpretation was the main goal?

Summary

- We use penalization to reduce the variance of our coefficients
- Ridge gives us stability, handles multicollinearity and is good for prediction
- Lasso induces sparsity (i.e. sets coefficients equal to zero), feature selection and is more interpretable